	#scenarios	#events	Max #events	#temporal links	Event length	%Non-linear	Domain
ProScrpt (Sakaguchi et al.)	2,077	7.46	9	6.95	4.64	39%	Daily
Schema-11 (Dror et al.)	11	7.91	11	7.18	3.48	27%	News
WikiHow Script (Lyu et al.)	2,991	8.37	20	7.37	9.63	0%	Daily

Table 2: Basic statistics of evaluation datasets. Max #events indicate the maximum number of events for a scenario. Event length is defined as the number of words in the event description. %Non-linear tells the proportion of temporal graphs that contain at least one branch. Two domains are considered, *Daily* activity and *News* journalism.

narratives in demonstrations to highlight the importance of narrative-aware few-shot demonstrations, and conduct a holistic evaluation of reference narratives in §5.2. We include a random baseline, where events are naively connected to form a *linear* temporal chain based on the order they appear in the input. We also experiment with two strong proprietary models, GPT-3.5[5] and GPT-4 (OpenAI, 2023)[6] to help gauge the gap between AI systems and human-level performance.

4.4 Evaluation Metrics

We denote the ground-truth and generated temporal graphs as $\mathcal{G}(\mathcal{V}, \mathcal{E})$ and $\hat{\mathcal{G}}(\mathcal{V}, \hat{\mathcal{E}})$, respectively. we compare both semantic and structural similarities between $\mathcal{G}$ and $\hat{\mathcal{G}}$, following prior work (Sakaguchi et al., 2021; Madaan et al., 2022). To evaluate semantic similarity, we report *precision (P)* and *recall (R)*, defined as below, as well as *F1*.

$$\text{Precision} = \frac{|\mathcal{E} \cap \hat{\mathcal{E}}|}{|\hat{\mathcal{E}}|} \quad \text{Recall} = \frac{|\mathcal{E} \cap \hat{\mathcal{E}}|}{|\mathcal{E}|}$$

To assess structural similarities, we consider:

- *Graph Edit Distance* (*GED*; Abu-Aisheh et al., 2015) calculates the minimum number of edits (node/edge removal/additions) to transform $\hat{\mathcal{G}}$ to a graph isomorphic to $\mathcal{G}$.
- *Graph Statistics*: fraction of the number of edges between $\hat{\mathcal{G}}$ and $\mathcal{G}$ ($\frac{|\hat{\mathcal{E}}|}{|\mathcal{E}|}$); the number of connected components in $\hat{\mathcal{G}}$, denoted as $k(\mathcal{G})$. The goal is to bring both statistics closer to 1, additionally ensuring $k(\mathcal{G})$ is at least 1.

We further calculate *Pair-wise Consistency* between $\hat{\mathcal{G}}_i$ and $\hat{\mathcal{G}}_j$, where we compare generated graphs, based on two randomly shuffled inputs, and compute the proportion of common temporal links produced in both graphs, i.e., $\frac{|\hat{\mathcal{E}}_i \cap \hat{\mathcal{E}}_j|}{|\hat{\mathcal{E}}_i \cup \hat{\mathcal{E}}_j|}$.

[5] https://chat.openai.com/; gpt-35-turbo-16k-0613, training data up to Sept. 2021.

[6] gpt-4-turbo-0125-preview, data up to Dec. 2023.

5 Results and Analyses

5.1 Main Results

Major results are included in Table 1, and the full results (across all 7 metrics) can be found in Table A1. Below are our major findings.

1) *With the few-shot setup, small LLMs are dramatically underperforming, reaching barely 50% of GPT-4's capabilities.* The three base models, whether using standard prompting or CoT, consistently under-perform GPT-4 and attain 40% to 60% of its average F1 scores. Among them, MISTRAL-7B achieves the highest F1 scores, while LLAMA3-8B produces temporal graphs most similar to the ground truth, as measured by GED.

2) *Unlike many other reasoning tasks, CoT does not always work for temporal reasoning and sometimes degrades performance.* Unlike mathematical or logical reasoning (Wei et al., 2022b), CoT prompting does not necessarily enhance model performance on temporal reasoning tasks. Across all three base models, there is a notable degradation in F1 and GED scores with CoT, except for LLAMA3's F1 scores. This is not TGG-specific, but rather a common pattern across various temporal understanding tasks (Chu et al., 2023), highlighting the need for specialized approaches to temporal reasoning. Outputs by CoT are included in Figure A6.

3) *GPT-4 is not always the champion, owing to the added safety layer.* GPT-4 implements safety measures through human-preference alignment (OpenAI, 2023), which enhances model safety by prompting more cautious responses, potentially leading to performance drop (Bai et al., 2022; Bekbayev et al., 2023). Especially on **Schema-11**, GPT-4 refrains from providing answers to sensitive scenarios like "bombing attacks",[7] and thus fails to produce a valid temporal graph.

4) *With* NOT*, small LLMs can perform comparably to GPT-3.5, or even take the lead.* When equipped with NOT, the overall semantic correctness (F1) and structural similarity (GED) of the

[7] In our experiments, we disabled content filtering.

	GEMMA-7B		MISTRAL-7B		LLAMA3-8B	
	F1↑	GED↓	F1↑	GED↓	F1↑	GED↓
NOT	21.8	2.48	35.4	2.14	39.5	1.87
FT	68.6	1.63	71.2	1.38	71.9	1.40

Table 3: Performance comparison between NOT and fine-tuning (FT) on ProScript. GPT-4 achieves 63.9 F1 and 1.64 GED, both lagging behind fine-tuned LLMs.

generated temporal graphs are significantly enhanced, regardless of which base LLM is used. The average improvement of F1 over naively prompting the base model is between 16% to 71%. As the power of the base LLM grows, NOT demonstrates greater consistency in its outputs. Notably, with LLAMA3-8B, the strongest base LLM, NOT achieves an F1 score that is comparable to GPT-3.5 (42.2 vs. 45.7), and even outperforms GPT-3.5/4 on GED. These results demonstrate the potential of applying NOT in a wide range of temporal understanding tasks in future research.

5) *Recounting temporally grounded narrative is a prerequisite for LLMs to generate temporal graphs accurately.* Without high-quality reference narratives, LLMs struggle to generate temporally grounded narratives, leading to a detrimental impact on NOT-augmented GEMMA-7B (e.g., a 0.7 F1 drop and a 0.67 GED increase).

6) *LLMs, including the powerful GPT-4, lag far behind human-level performance in temporal reasoning.* The SOTA F1 score (by GPT-4) on ProScript is 63.9, whereas the human baseline F1 is 89.3 (Sakaguchi et al., 2021). While NOT has notably narrowed the gap between small and large LLMs, AI models have not mastered temporal reasoning yet, and further research efforts are needed for LLMs to match human performance.

Comparison with fine-tuned LLMs. To evaluate the performance gap between the NOT prompting technique and the computational-intense fine-tuning (FT) approach, we conduct a side experiment on the ProScript dataset. Specifically, each instruction-tuned base LLM is fine-tuned on the ProScript training set, utilizing LoRA (Hu et al., 2022) and mixed-precision training. We follow the same setting as in §4.2 where each training example is prepended with 5-shot demonstrations. While significant performance disparities between NOT and FT are observed across the board, the narrowing gap suggests the growing potential of NOT as the underlying LLM continues to evolve. Moreover, fine-tuned small LLMs consistently outperform the few-shot GPT-4, which is the best-performing generalist model on the ProScript dataset. This underscores the continued efficacy of FT in building specialized models, even in the era of LLMs.

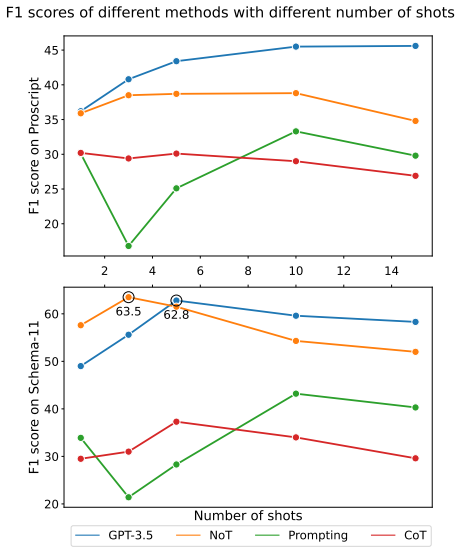

Figure 3: F1 scores on ProScript and Schema-11 in relation to the number of shots in demonstrations. We identify the **instability** in the standard prompting, and the **performance plateau** after 5 shots.

5.2 Further Studies on NOT

We conduct ablation studies using LLAMA3-8B, to explore the effect of the few-shot demonstrations and the recounted reference narratives.

Does the number of shots matter? Figure 3 illustrates how F1 scores change with the number of shots in demonstrations. As can be seen, GPT-3.5 and NOT show resilience to changes in shot numbers after an initial sharp increase. The performance nearly stabilizes in the range of 5-10 shots, though a slight drop is observed later, presumably due to insufficient capability of long-context comprehension (Liu et al., 2023; Li et al., 2024). Of particular interest is the performance of NOT with 3 shots on Schema-11, outperforming the best variant of GPT-3.5 (F1 of 63.5 vs. 62.8). This further illustrates NOT's potential of boosting small LLMs in the long run. It is also noticeable that F1 scores of the standard prompting technique have a V-shape between 1-shot and 5-shot, highlighting its sensitiveness to in-context demonstrations.

We also display the GED scores in relation to number of shots in Figure A1. We observe similar instability in the standard prompting technique, along with the performance plateau after 5 shots.

What characteristics define effective reference narratives? Given that reference narratives in

NoT are machine-generated, we aim to explore what qualities matter most for the TGG task. Here, the three variables influencing reference narratives are: (1) narrative generation model (GPT-3.5 vs. GPT-4), (2) input format (alphabetical vs. descriptive), and (3) 4 meta prompt types (varying degrees of factuality and readability). We show detailed meta prompts in Appendix D.

Figure 4 and Figure A2 show results of F1 and GED with varying meta prompts. Surprisingly, the choice of the generator does not significantly impact the graph quality, with average F1 scores of 36.4 for GPT-3.5 and 37.0 for GPT-4, and GED scores of 1.90 vs. 1.94. Similarly, there is no significant difference between alphabetical and descriptive input formats. The most *impactful* factor is the meta prompt type. Grouping performance bars by prompt type reveals a clear variance in model performance. Among the first three groups, *Simple English* narratives, i.e., good for 10-year-olds, stand out. This suggests that narratives should be simple and concise, as verbose ones are less effective. We find that *News Report* narratives prioritize procedural and factual content, minimizing distractions like descriptive settings or figurative language that can often be found in both fiction or non-fiction stories. We thus combine *Simple English* and *News Report* to leverage their strengths, dubbed *Simple Report*. In summary, we identify three key characteristics for quality reference narratives: *conciseness*, *simplicity* and *factuality*.

How faithful is the temporal graph to intermediate narratives? Here, we look into whether NoT-augmented LLMs are **self-faithful**, i.e., whether the narrative and the temporal graph **align** in terms of the temporal order of events. Higher self-faithfulness is crucial and desired, as misalignment would diminish the effort of generating a temporally grounded narrative.[8]

Motivated by the recent success of using LLMs as judges (Zheng et al., 2023; Zhang et al., 2024a), we employ GPT-4 to assess the self-faithfulness of 600 randomly sampled outputs by NoT-augmented Llama3-8B. We prompt GPT-4 to perform a 5-way assessment and provide judgment rationales. Additionally, GPT-4 is instructed to count the temporal links in the temporal graphs and identify aligned temporal links for a sanity check. This helps humans capture the failure modes and make necessary interventions. Based on automated responses and on-demand human inspections, we find a medium-to-high alignment of 72.8%. Details of templates and the inspection process are included in Appendix F.

[8]Faithfulness $\neq$ correctness. A faithful temporal graph may still contain logical errors from the generated narratives.

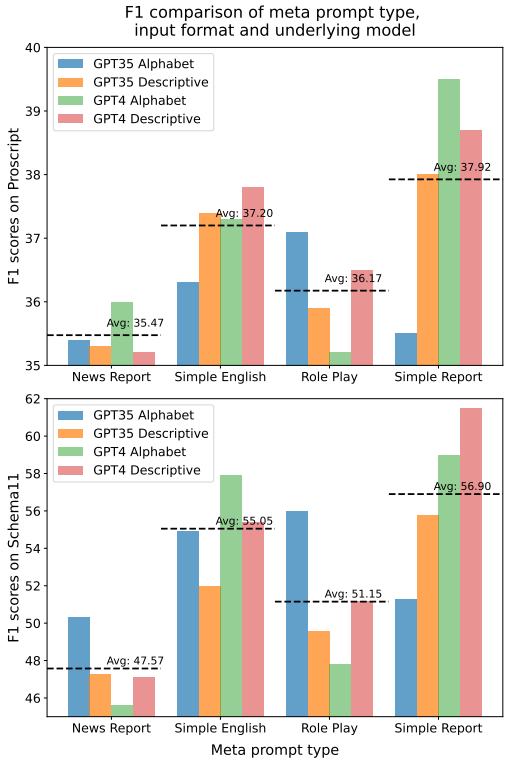

Figure 4: F1 scores on ProScript and Schema-11 with different meta prompts. Average performance grouped by prompt type is also shown. Notably, using a *Simple Report*-style, GPT-4 generated narratives lead to the best score due to its **conciseness**, **simplicity** and **factuality**, which are essential qualities for a *high-quality* reference narrative.

6 Conclusion

In this paper, we assess the inherent, global temporal reasoning capabilities of LLMs, by studying the core challenge of temporal reasoning—temporal graph generation (TGG). To this end, we propose Narrative-of-Thought (NoT), a novel prompting technique tailored for temporal reasoning. Concretely, with few-show narrative-aware demonstrations as references, NoT prompts LLMs to first generate a temporally grounded narrative and then sort the input events topologically into a temporal graph, by manipulating the generation in code space. Extensive experiments showcase NoT's effectiveness, demonstrated by its superior performance over GPT-3.5 on multiple metrics, as well as the compatibility of NoT with various LLMs.